IAM-02: SSO and Authentication Configuration

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Setting Up Enterprise Authentication

Single Sign-On (SSO) is the foundation of enterprise IAM. Configure SSO before setting up groups, policies, or user provisioning. This notebook covers SAML configuration, IdP integration, and authentication best practices.

Table of Contents

1. Why SSO First?
2. SAML 2.0 Fundamentals
3. IdP Configuration
4. Dynatrace SAML Setup
5. Attribute Mapping
6. Testing SSO
7. Multi-Factor Authentication
8. Troubleshooting SSO

Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Gen3 IAM enabled
Permissions	Account administrator
IdP Access	Admin access to enterprise IdP (Okta, Azure AD, etc.)

1. Why SSO First?

SSO must be configured before other IAM components because:

Dependency	Reason
Groups	Groups are synced from IdP via SAML assertions
Users	Users are provisioned through SSO (SCIM or JIT)
Policies	Policies are assigned to groups that come from IdP
Audit	Authentication events require SSO integration

Authentication Flow

```
User → IdP Login → SAML Assertion → Dynatrace
                          ↓
               Contains: Email, Groups, Attributes
                          ↓
               Dynatrace maps to local groups
                          ↓
               Policies and boundaries applied
```

Order of Operations

1. **Configure SSO** (this notebook)
2. Design group architecture (IAM-03)
3. Create policies (IAM-04)
4. Design boundaries (IAM-05)

5. Enable SCIM/JIT provisioning (IAM-06)

2. SAML 2.0 Fundamentals

Dynatrace uses SAML 2.0 for SSO authentication.

Key Concepts

Term	Definition
IdP	Identity Provider - authenticates users (Okta, Azure AD, etc.)
SP	Service Provider - Dynatrace, the application being accessed
Assertion	XML document containing user identity and attributes
ACS URL	Assertion Consumer Service - where Dynatrace receives assertions
Entity ID	Unique identifier for each party

SAML Flow

```
1. User accesses Dynatrace
2. Dynatrace redirects to IdP
3. User authenticates with IdP
4. IdP generates SAML assertion
5. IdP redirects back to Dynatrace with assertion
6. Dynatrace validates assertion
7. User is logged in with appropriate permissions
```

Required SAML Attributes

Attribute	Purpose	Example
NameID	User identifier (email)	`user@company.com`
Groups	Group memberships	`IT-DT-Admins, All-Employees`
FirstName	Display name	`John`
LastName	Display name	`Smith`

3. IdP Configuration

Configure your Identity Provider to work with Dynatrace.

Supported Identity Providers

IdP	SSO Support	SCIM Setup Documented by Dynatrace	Notes
Okta	Full	Yes	Native integration
Azure AD (Entra ID)	Full	Yes	Enterprise app
OneLogin	Full	No	App catalog
PingFederate	Full	No	Federation server
Google Workspace	Full	No	Custom SAML app
ADFS	Full	No	On-premises

Dynatrace's SCIM page documents setup procedures for Azure and Okta only. In community practice, other IdPs with a generic SCIM 2.0 client are configured against the same endpoint — verify with your IdP vendor and test before relying on it; otherwise provision those users through SAML just-in-time creation.

> **Sources:** SCIM (DT docs) — *"To continue integrating Dynatrace SCIM with your IdP, select the procedure appropriate for your IdP:"* (Azure, Okta); SAML (DT docs) — *"Just-in-Time provisioning is supported."*

IdP Setup Steps (Generic)

1. **Create SAML Application**
   - Log into IdP admin console
   - Create new SAML 2.0 application
   - Name it "Dynatrace" or "Dynatrace [Environment]"

2. **Configure Service Provider Settings**
   - Take the Entity ID and ACS URL from the Dynatrace SP metadata. They depend on the federation type you choose in Section 4:
   - **Global federation:** Entity ID (audience) `https://sso.dynatrace.com:443/saml2/login`; ACS / Single Sign On URL `https://sso.dynatrace.com:443/saml2/sp/consumer`
   - **Account or environment federation:** download the unique SP metadata while creating the SAML configuration. The ACS URL has the form `https://sso.dynatrace.com/identity-federation/sp/consumer/account/<account-UUID>/federation/<configuration-UUID>`
   - Name ID Format: Email address
3. **Configure Attributes**
   - Map user email to NameID
   - Add groups attribute
   - Add first/last name attributes
4. **Assign Users/Groups**
   - Assign IdP groups that should access Dynatrace
   - Or assign all users and control access via Dynatrace groups

> **Sources:** SAML (DT docs) — *"https://sso.dynatrace.com:443/saml2/login for Entity ID (Audience Restriction).", "https://sso.dynatrace.com:443/saml2/sp/consumer for Single Sign On URL, Destination URL, Recipient URL."*

Okta Configuration Example

1. **Create Application**
   - Applications → Create App Integration
   - Select SAML 2.0
   - Name: "Dynatrace"
2. **SAML Settings**

`Single sign-on URL: <ACS URL from the Dynatrace SP metadata> (global federation: https://sso.dynatrace.com:443/saml2/sp/consumer) Audience URI (SP Entity ID): <Entity ID from the Dynatrace SP metadata> (global`

`federation: https://sso.dynatrace.com:443/saml2/login) Name ID format: EmailAddress Application username: Email`

3. **Attribute Statements**
Name	Value
`email`	`user.email`
`firstName`	`user.firstName`
`lastName`	`user.lastName`

4. **Group Attribute Statements**
Name	Filter
`groups`	Matches regex `.*` (or filter to DT groups)

Azure AD (Entra ID) Configuration Example

1. **Enterprise Application**
   - Enterprise Applications → New Application
   - Create your own application
   - Name: "Dynatrace"

2. **Single Sign-On → SAML**
   `Identifier (Entity ID): <Entity ID from the Dynatrace SP metadata> (global federation: https://sso.dynatrace.com:443/saml2/login) Reply URL (ACS URL): <ACS URL from the Dynatrace SP metadata> (global federation: https://sso.dynatrace.com:443/saml2/sp/consumer)`

3. **Attributes & Claims**
Claim	Source attribute
`emailaddress`	`user.mail`
`givenname`	`user.givenname`
`surname`	`user.surname`
`groups`	Group IDs or names

4. Dynatrace SAML Setup

Configure Dynatrace to accept SAML assertions from your IdP.

Steps in Dynatrace Account Management

1. **Create a fallback user**
   - A non-federated user account in a group with the *View and manage users and groups* permission, so you can still sign in if the SAML configuration fails

2. **Verify domain ownership**
   - Identity & access management → Domain verification — add your email domain, publish the DNS TXT record, and verify it

3. **Create the SAML configuration**
   - Identity & access management → SAML configuration → New configuration
   - Choose the federation type — global, account, or environment

4. **Enter IdP Information**
   - **Name**: Descriptive name (e.g., "Corporate Okta")
   - **IdP Entity ID**: From IdP metadata
   - **IdP SSO URL**: From IdP metadata
   - **IdP Certificate**: Upload X.509 certificate from IdP

5. **Get Dynatrace SP Information**
   - Copy ACS URL and Entity ID to configure in IdP
   - Download Dynatrace SP metadata if IdP supports it

> **Sources:** SAML (DT docs) — *"Before you can configure the domain for which you want to set up SAML, you need to prove ownership of the domain."*, *"go to Identity & access management > SAML configuration and select New configuration"*.

Configuration Options

Option	Recommendation	Notes
Sign the whole SAML response	Required	Signing only the assertion is not enough — the IdP must sign the entire message
Encrypt assertions	Not supported	Leave assertion encryption off in the IdP
Sign requests	Optional	SP signs auth requests
Allow IdP-initiated	Optional	Supported, with RelayState

> **Sources:** SAML (DT docs) — *"The entire SAML message must be signed"*, *"No assertion encryption."*, *"Yes, both IdP-initiated sign-in and RelayState are supported."*

5. Attribute Mapping

Map SAML attributes to Dynatrace user properties.

User Attribute Mapping

SAML Attribute	Dynatrace Field	Required
NameID	User email/login	Yes
firstName	First name	Recommended
lastName	Last name	Recommended
email	Email (if not NameID)	If NameID is different

Group Attribute Mapping

Group values from SAML assertions map to Dynatrace groups:

```
IdP Group (in assertion)  →  Dynatrace Group
IT-DT-Platform-Admins     →  dt-platform-admins
IT-DT-Checkout-Team       →  dt-checkout-editors
IT-DT-Security-Audit      →  dt-security-viewers
```

Configuring Group Mapping

Mapping is configured on each Dynatrace group, not on the SAML configuration:

1. **Open the group** — Account Management → **Identity & access management** → **Group management**, then the group's **Edit** pane. Test with a new group first.
2. **Set the SAML Group Attribute Value** — one or more federated group values your IdP sends in the groups attribute. This is the value in the assertion, which is typically an identifier such as an LDAP ID rather than the group's display name.

3. **Save, then test** — sign in from a new incognito window before signing out of your current session.

> ⚠️ **Converting an existing group wipes its members.** Adding SAML Group Attribute Values to a LOCAL group turns it into a SAML group and removes every user currently assigned to it. Plan the switch for each group, and make sure the users' IdP groups are sending the right values first.
>
> **Sources:** SAML (DT docs) — *"This typically isn't a group display name."*, *"Switching a LOCAL group to SAML removes all user assignments to that group."*

Group Naming Strategy

Approach	IdP Group	DT Group	Pros/Cons
Exact match	`dt-admins`	`dt-admins`	Simple, but only where the IdP sends group names rather than IDs
Prefixed	`IT-DT-Admins`	`dt-admins`	Explicit mapping needed
Descriptive	`Dynatrace Platform Admins`	`dt-platform-admins`	Clear in IdP, mapping required

6. Testing SSO

Verify SSO works correctly before rolling out.

Test Checklist

```
☐ SP-initiated login works (start from Dynatrace)
☐ IdP-initiated login works (start from IdP portal)
☐ User attributes map correctly (name, email)
☐ Groups sync correctly
☐ Logout works (single logout)
☐ Session timeout behaves correctly
```

Testing Steps

1. **Test with Admin Account First**
   - Use an IdP admin account
   - Verify login works
   - Check group assignments
2. **Test SP-Initiated Login**
   - Go to Dynatrace login page
   - Click SSO login
   - Should redirect to IdP
   - Authenticate and return to Dynatrace
3. **Test IdP-Initiated Login**
   - Go to IdP portal
   - Click Dynatrace app tile
   - Should land in Dynatrace authenticated
4. **Verify User Profile**
   - Check user's name is correct
   - Check user's email is correct
   - Check group memberships

Common Test Failures

Symptom	Likely Cause	Fix
Redirect loop	Misconfigured ACS URL	Verify URL matches exactly
Invalid signature	Certificate mismatch	Re-upload IdP certificate
User not found	NameID format wrong	Use email format
No groups	Groups attribute missing	Add groups to assertion

7. Multi-Factor Authentication

MFA must be enforced at the IdP for Dynatrace access — Dynatrace does not support configuring MFA itself.

> **Sources:** SAML (DT docs) — *"Configuring multi-factor authentication (MFA) is not supported by Dynatrace."*

MFA Strategy

Approach	Description	Recommendation
IdP MFA	MFA at IdP login	Recommended
Conditional	MFA based on risk	For adaptive security

IdP MFA Configuration

Okta:
- Security → Authenticators
- Add factors (TOTP, Push, WebAuthn)
- Create sign-on policy requiring MFA for Dynatrace app

Azure AD:
- Azure AD → Security → MFA
- Conditional Access policies
- Require MFA for Dynatrace enterprise app

MFA Best Practices

Practice	Rationale
Require MFA for all users	Baseline security
Use phishing-resistant factors	WebAuthn > TOTP > SMS
Allow backup factors	Prevent lockout
Enforce for privileged users	Critical for admins

8. Troubleshooting SSO

Common SSO issues and their resolutions.

SAML Debugging Tools

Tool	Purpose
Browser dev tools	View SAML response in network tab
SAML tracer extension	Decode SAML in browser
Base64 decoder	Decode assertion manually
IdP logs	View authentication events

Decoding SAML Response

1. Open browser dev tools → Network
2. Perform SSO login
3. Find POST to ACS URL
4. Copy SAMLResponse value
5. Base64 decode to see XML

Common Issues

Issue	Symptom	Resolution
Clock skew	"Assertion expired"	Sync IdP/Dynatrace time
Certificate expired	Signature invalid	Renew and update cert
Wrong ACS URL	404 or loop	Correct URL in IdP
Missing groups	No permissions	Add groups to assertion
Wrong NameID	User mismatch	Use email format
Audience mismatch	Audience error	Match Entity ID exactly

```
// Authenticated API calls by credential type and grant type (24 h)
// Retitled 10/12/2026: authentication.type is set on every audited request,
so this is API traffic, not
// SSO sign-ins. Sign-ins are event.type == "LOGIN" (see IAM-07 § 2).
// Data object corrected 08/12/2026. The Dynatrace audit trail is NOT in
`logs`: this cell used
// `fetch logs | filter matchesPhrase(log.source, "audit")`, and no log.source
on a Grail tenant
// contains "audit" — the filter matched nothing, silently, forever. Platform
audit records live in
// `dt.system.events` with `event.kind == "AUDIT_EVENT"` (265,000+ records
over 7 days here), and
// they are STRUCTURED, so the old `matchesPhrase(content, ...)` string-
scraping is replaced by real
// field predicates. Key fields: user.id, user.organization, event.type
(GET/POST/PUT/PATCH/DELETE/
// CREATE/LOGIN/app.opened), event.outcome (HTTP status or "success"),
authentication.type
// (OAUTH2/TOKEN/NONE), authentication.grant.type, resource (the API path),
origin.address,
// origin.type, request.source, dt.app.id, event.provider.
// Enumerate your own with:
//   fetch dt.system.events, from:-24h | filter event.kind == "AUDIT_EVENT" |
limit 1
fetch dt.system.events, from:-24h
| filter event.kind == "AUDIT_EVENT"
| filter isNotNull(authentication.type)
| summarize events = count(), by:{authentication.type,
authentication.grant.type, event.outcome}
| sort events desc
```

```
// Find failed sign-in attempts (event.type LOGIN with outcome "failure" —
other non-2xx outcomes
// are failed API calls, and 304 Not Modified is a success)
// Data object corrected 08/12/2026. The Dynatrace audit trail is NOT in
`logs`: this cell used
// `fetch logs | filter matchesPhrase(log.source, "audit")`, and no log.source
on a Grail tenant
// contains "audit" — the filter matched nothing, silently, forever. Platform
audit records live in
// `dt.system.events` with `event.kind == "AUDIT_EVENT"` (265,000+ records
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scraping is replaced by real
// field predicates. Key fields: user.id, user.organization, event.type
(GET/POST/PUT/PATCH/DELETE/
// CREATE/LOGIN/app.opened), event.outcome (HTTP status or "success"),
authentication.type
// (OAUTH2/TOKEN/NONE), authentication.grant.type, resource (the API path),
origin.address,
// origin.type, request.source, dt.app.id, event.provider.
// Enumerate your own with:
//   fetch dt.system.events, from:-24h | filter event.kind == "AUDIT_EVENT" |
limit 1
fetch dt.system.events, from:-7d
| filter event.kind == "AUDIT_EVENT"
| filter event.type == "LOGIN" and event.outcome == "failure"
| fields timestamp, user.id, authentication.type, event.outcome,
origin.address
| sort timestamp desc
| limit 50
```

Next Steps

With SSO configured, proceed to group architecture:

Recommended Path

1. **IAM-03: Group Architecture and Design** - Design groups that map from IdP
2. **IAM-04: Policy Authoring** - Create policies for groups
3. **IAM-06: User Lifecycle** - Enable SCIM/JIT provisioning

SSO Checklist

Before moving on, ensure you have:

- [ ] IdP SAML application configured
- [ ] Dynatrace SAML configuration created
- [ ] Attribute mapping configured
- [ ] SAML Group Attribute Values planned per group
- [ ] SSO login tested successfully
- [ ] MFA enabled at IdP

Summary

In this notebook, you learned:

- Why SSO must be configured first
- SAML 2.0 fundamentals
- IdP configuration for Okta and Azure AD
- Dynatrace SAML setup process
- Attribute and group mapping
- SSO testing methodology
- MFA best practices
- Troubleshooting common SSO issues

References

- Single Sign-On
- SAML 2.0 Configuration
- Okta Integration
- Azure AD Integration

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